In [1]:
```
import pandas as pd
```

In [3]:
```
df = pd.read_csv(r"C:\Users\abhis\Downloads\Salary_Data.csv")
df
```

Out[3]:
<style scoped> .dataframe tbody tr th:only-of-type { vertical-align: middle; } .dataframe tbody tr th { vertical-align: top; } .dataframe thead th { text-align: right; } </style> <thead> <th></th> <th>YearsExperience</th> <th>Salary</th> </thead> <tbody> <th>0</th> <th>1</th> <th>2</th> <th>3</th> <th>4</th> <th>5</th> <th>6</th> <th>7</th> <th>8</th> <th>9</th> <th>10</th> <th>11</th> <th>12</th> <th>13</th> <th>14</th> <th>15</th> <th>16</th> <th>17</th> <th>18</th> <th>19</th> <th>20</th> <th>21</th> <th>22</th> <th>23</th> <th>24</th> <th>25</th> <th>26</th> <th>27</th> <th>28</th> <th>29</th> </tbody> <tbody> </tbody>

| | |
|---|---|
| 1.1 | 39343.0 |
| 1.3 | 46205.0 |
| 1.5 | 37731.0 |
| 2.0 | 43525.0 |
| 2.2 | 39891.0 |
| 2.9 | 56642.0 |
| 3.0 | 60150.0 |
| 3.2 | 54445.0 |
| 3.2 | 64445.0 |
| 3.7 | 57189.0 |
| 3.9 | 63218.0 |
| 4.0 | 55794.0 |
| 4.0 | 56957.0 |
| 4.1 | 57081.0 |
| 4.5 | 61111.0 |
| 4.9 | 67938.0 |
| 5.1 | 66029.0 |
| 5.3 | 83088.0 |
| 5.9 | 81363.0 |
| 6.0 | 93940.0 |
| 6.8 | 91738.0 |
| 7.1 | 98273.0 |
| 7.9 | 101302.0 |
| 8.2 | 113812.0 |
| 8.7 | 109431.0 |
| 9.0 | 105582.0 |
| 9.5 | 116969.0 |
| 9.6 | 112635.0 |
| 10.3 | 122391.0 |
| 10.5 | 121872.0 |

In [4]:
```
import pandas as pd
df = pd.read_csv(r"C:\Users\abhis\Downloads\Salary_Data.csv")
print(df.head())
print(df.tail())
print(df.info())
print(df.describe())
print(df.shape)

df
```

```
   YearsExperience   Salary
0              1.1  39343.0
1              1.3  46205.0
2              1.5  37731.0
3              2.0  43525.0
4              2.2  39891.0
    YearsExperience    Salary
25              9.0  105582.0
26              9.5  116969.0
27              9.6  112635.0
28             10.3  122391.0
29             10.5  121872.0
<class 'pandas.DataFrame'>
RangeIndex: 30 entries, 0 to 29
Data columns (total 2 columns):
 #   Column           Non-Null Count  Dtype  
---  ------           --------------  -----  
 0   YearsExperience  30 non-null     float64
 1   Salary           30 non-null     float64
dtypes: float64(2)
memory usage: 612.0 bytes
None
       YearsExperience         Salary
count        30.000000      30.000000
mean          5.313333   76003.000000
std           2.837888   27414.429785
min           1.100000   37731.000000
25%           3.200000   56720.750000
50%           4.700000   65237.000000
75%           7.700000  100544.750000
max          10.500000  122391.000000
(30, 2)
```

Out[4]: <style scoped> .dataframe tbody tr th:only-of-type { vertical-align: middle; } .dataframe tbody tr th { vertical-align: top; } .dataframe thead th { text-align: right; } </style> <thead> <th></th> <th>YearsExperience</th> <th>Salary</th> </thead> <tbody> <th>0</th> <th>1</th> <th>2</th> <th>3</th> <th>4</th> <th>5</th> <th>6</th> <th>7</th> <th>8</th> <th>9</th> <th>10</th> <th>11</th> <th>12</th> <th>13</th> <th>14</th> <th>15</th> <th>16</th> <th>17</th> <th>18</th> <th>19</th> <th>20</th> <th>21</th> <th>22</th> <th>23</th> <th>24</th> <th>25</th> <th>26</th> <th>27</th> <th>28</th> <th>29</th> </tbody> <tbody> </tbody>

| | |
|---|---|
| 1.1 | 39343.0 |
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| 2.0 | 43525.0 |
| 2.2 | 39891.0 |
| 2.9 | 56642.0 |
| 3.0 | 60150.0 |
| 3.2 | 54445.0 |
| 3.2 | 64445.0 |
| 3.7 | 57189.0 |
| 3.9 | 63218.0 |
| 4.0 | 55794.0 |
| 4.0 | 56957.0 |
| 4.1 | 57081.0 |
| 4.5 | 61111.0 |
| 4.9 | 67938.0 |
| 5.1 | 66029.0 |
| 5.3 | 83088.0 |
| 5.9 | 81363.0 |
| 6.0 | 93940.0 |
| 6.8 | 91738.0 |
| 7.1 | 98273.0 |
| 7.9 | 101302.0 |
| 8.2 | 113812.0 |
| 8.7 | 109431.0 |
| 9.0 | 105582.0 |
| 9.5 | 116969.0 |
| 9.6 | 112635.0 |
| 10.3 | 122391.0 |
| 10.5 | 121872.0 |

In [6]:
```
x=df[['YearsExperience']]
y=df[['Salary']]
print(x)
print(y)
```

```
    YearsExperience
0               1.1
1               1.3
2               1.5
3               2.0
4               2.2
5               2.9
6               3.0
7               3.2
8               3.2
9               3.7
10              3.9
11              4.0
12              4.0
13              4.1
14              4.5
15              4.9
16              5.1
17              5.3
18              5.9
19              6.0
20              6.8
21              7.1
22              7.9
23              8.2
24              8.7
25              9.0
26              9.5
27              9.6
28             10.3
29             10.5
      Salary
0    39343.0
1    46205.0
2    37731.0
3    43525.0
4    39891.0
5    56642.0
6    60150.0
7    54445.0
8    64445.0
9    57189.0
10   63218.0
11   55794.0
12   56957.0
13   57081.0
14   61111.0
15   67938.0
16   66029.0
17   83088.0
18   81363.0
19   93940.0
20   91738.0
21   98273.0
22  101302.0
23  113812.0
24  109431.0
25  105582.0
26  116969.0
27  112635.0
28  122391.0
29  121872.0
```

In [12]:
```
pip install scikit-learn
```

```
Requirement already satisfied: scikit-learn in C:\Users\abhis\AppData\Local\Programs\Python\Python314\Lib\site-p
ackages (1.9.1)
Requirement already satisfied: numpy>=1.24.1 in C:\Users\abhis\AppData\Local\Programs\Python\Python314\Lib\site-
packages (from scikit-learn) (2.5.3)
Requirement already satisfied: scipy>=1.10.0 in C:\Users\abhis\AppData\Local\Programs\Python\Python314\Lib\site-
packages (from scikit-learn) (1.18.1)
Requirement already satisfied: joblib>=1.4.0 in C:\Users\abhis\AppData\Local\Programs\Python\Python314\Lib\site-
packages (from scikit-learn) (1.6.0)
Requirement already satisfied: narwhals>=2.0.1 in C:\Users\abhis\AppData\Local\Programs\Python\Python314\Lib\sit
e-packages (from scikit-learn) (2.26.0)
Requirement already satisfied: threadpoolctl>=3.5.0 in C:\Users\abhis\AppData\Local\Programs\Python\Python314\Li
b\site-packages (from scikit-learn) (3.7.0)
Requirement already satisfied: cloudpickle>=3.0 in C:\Users\abhis\AppData\Local\Programs\Python\Python314\Lib\si
te-packages (from joblib>=1.4.0->scikit-learn) (3.1.2)
Note: you may need to restart the kernel to use updated packages.
```

In [14]:
```
from sklearn.model_selection import train_test_split
x_train,x_test,y_train,y_test=train_test_split(x,y,test_size=0.2,random_state=12)
```

In [15]:
```
from sklearn.linear_model import LinearRegression
slr=LinearRegression()
slr.fit(x_train,y_train)
slr.predict(x_test)
```

Out[15]:
```
array([[ 55675.5913984 ],
       [100372.57413579],
       [ 81352.58148158],
       [ 68038.58662364],
       [ 37606.59837691],
       [ 92764.5770741 ]])
```

In [4]:
```
pip install matplotlib
```

```
Collecting matplotlib
  Downloading matplotlib-3.11.2-cp314-cp314-win_amd64.whl.metadata (80 kB)
Collecting contourpy>=1.0.1 (from matplotlib)
  Downloading contourpy-1.4.0-cp314-cp314-win_amd64.whl.metadata (3.8 kB)
Collecting cycler>=0.10 (from matplotlib)
  Downloading cycler-0.12.1-py3-none-any.whl.metadata (3.8 kB)
Collecting fonttools>=4.28.2 (from matplotlib)
  Downloading fonttools-4.66.1-cp314-cp314-win_amd64.whl.metadata (132 kB)
Collecting kiwisolver>=1.3.1 (from matplotlib)
  Downloading kiwisolver-1.5.1-cp314-cp314-win_amd64.whl.metadata (5.2 kB)
Requirement already satisfied: numpy>=1.25 in C:\Users\abhis\AppData\Local\Programs\Python\Python314\Lib\site-pa
ckages (from matplotlib) (2.5.3)
Requirement already satisfied: packaging>=20.0 in C:\Users\abhis\AppData\Local\Programs\Python\Python314\Lib\sit
e-packages (from matplotlib) (26.3)
Collecting pillow>=9 (from matplotlib)
  Downloading pillow-12.3.0-cp314-cp314-win_amd64.whl.metadata (9.3 kB)
Collecting pyparsing>=3 (from matplotlib)
  Downloading pyparsing-3.3.3-py3-none-any.whl.metadata (5.9 kB)
Requirement already satisfied: python-dateutil>=2.7 in C:\Users\abhis\AppData\Local\Programs\Python\Python314\Li
b\site-packages (from matplotlib) (2.9.0.post0)
Requirement already satisfied: six>=1.5 in C:\Users\abhis\AppData\Local\Programs\Python\Python314\Lib\site-packa
ges (from python-dateutil>=2.7->matplotlib) (1.17.0)
Downloading matplotlib-3.11.2-cp314-cp314-win_amd64.whl (9.5 MB)
   ---------------------------------------- 0.0/9.5 MB ? eta -:--:--
   -------- ------------------------------- 2.1/9.5 MB 11.8 MB/s eta 0:00:01
   ------------------------- -------------- 6.3/9.5 MB 17.0 MB/s eta 0:00:01
   ---------------------------------------  9.4/9.5 MB 17.3 MB/s eta 0:00:01
   ---------------------------------------- 9.5/9.5 MB 15.5 MB/s  0:00:00
Downloading contourpy-1.4.0-cp314-cp314-win_amd64.whl (240 kB)
Downloading cycler-0.12.1-py3-none-any.whl (8.3 kB)
Downloading fonttools-4.66.1-cp314-cp314-win_amd64.whl (2.5 MB)
   ---------------------------------------- 0.0/2.5 MB ? eta -:--:--
   ---------------------------------------- 2.5/2.5 MB 14.4 MB/s  0:00:00
Downloading kiwisolver-1.5.1-cp314-cp314-win_amd64.whl (72 kB)
Downloading pillow-12.3.0-cp314-cp314-win_amd64.whl (7.2 MB)
   ---------------------------------------- 0.0/7.2 MB ? eta -:--:--
   -------------- ------------------------ 2.9/7.2 MB 13.9 MB/s eta 0:00:01
   -------------------- ------------------- 3.7/7.2 MB 10.6 MB/s eta 0:00:01
   ------------------------- ------------ 5.0/7.2 MB 7.7 MB/s eta 0:00:01
   ---------------------------------- ----- 6.3/7.2 MB 8.3 MB/s eta 0:00:01
   ---------------------------------------- 7.2/7.2 MB 6.7 MB/s  0:00:01
Downloading pyparsing-3.3.3-py3-none-any.whl (126 kB)
Installing collected packages: pyparsing, pillow, kiwisolver, fonttools, cycler, contourpy, matplotlib

   ---------------------------------------- 0/7 [pyparsing]
   ---------------------------------------- 0/7 [pyparsing]
   ----- ---------------------------------- 1/7 [pillow]
   ----- ---------------------------------- 1/7 [pillow]
   ----- ---------------------------------- 1/7 [pillow]
   ----- ---------------------------------- 1/7 [pillow]
   ----- ---------------------------------- 1/7 [pillow]
   ----- ---------------------------------- 1/7 [pillow]
   ----- ---------------------------------- 1/7 [pillow]
   ----- ---------------------------------- 1/7 [pillow]
   ----- ---------------------------------- 1/7 [pillow]
   ----- ---------------------------------- 1/7 [pillow]
   ----- ---------------------------------- 1/7 [pillow]
   ----- ---------------------------------- 1/7 [pillow]
   ----- ---------------------------------- 1/7 [pillow]
   ----------------- ---------------------- 3/7 [fonttools]
   ----------------- ---------------------- 3/7 [fonttools]
   ----------------- ---------------------- 3/7 [fonttools]
   ----------------- ---------------------- 3/7 [fonttools]
   ----------------- ---------------------- 3/7 [fonttools]
   ----------------- ---------------------- 3/7 [fonttools]
   ----------------- ---------------------- 3/7 [fonttools]
   ----------------- ---------------------- 3/7 [fonttools]
   ----------------- ---------------------- 3/7 [fonttools]
   ----------------- ---------------------- 3/7 [fonttools]
   ----------------- ---------------------- 3/7 [fonttools]
```

---------------- ---------------------- 3/7 [fonttools]
---------------- ---------------------- 3/7 [fonttools]
---------------- ---------------------- 3/7 [fonttools]

```
   ---------------------------------- ----- 6/7 [matplotlib]
   ---------------------------------- ----- 6/7 [matplotlib]
   ---------------------------------- ----- 6/7 [matplotlib]
   ---------------------------------- ----- 6/7 [matplotlib]
   ---------------------------------- ----- 6/7 [matplotlib]
   ---------------------------------- ----- 6/7 [matplotlib]
   ---------------------------------------- 7/7 [matplotlib]

Successfully installed contourpy-1.4.0 cycler-0.12.1 fonttools-4.66.1 kiwisolver-1.5.1 matplotlib-3.11.2 pillow-
12.3.0 pyparsing-3.3.3
Note: you may need to restart the kernel to use updated packages.
```

In [5]:
```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_absolute_error, mean_squared_error, r2_score
```

In [7]:
```
df=pd.read_csv(r"C:\Users\abhis\Downloads\Salary_Data.csv")
print("First 5 Rows:")
print(df.head())
print("\nLast 5 Rows:")
print(df.tail())
```

```
First 5 Rows:
   YearsExperience   Salary
0              1.1  39343.0
1              1.3  46205.0
2              1.5  37731.0
3              2.0  43525.0
4              2.2  39891.0

Last 5 Rows:
    YearsExperience    Salary
25              9.0  105582.0
26              9.5  116969.0
27              9.6  112635.0
28             10.3  122391.0
29             10.5  121872.0
```

In [8]:
```
print("\nNumerical Columns:")
print(df.select_dtypes(include=np.number).columns)
print("\nCategorical Columns:")
print(df.select_dtypes(include='object').columns)
```

```
Numerical Columns:
Index(['YearsExperience', 'Salary'], dtype='str')

Categorical Columns:
Index([], dtype='str')
```

In [11]:
```
df = df.drop_duplicates()
```

In [12]:
```
for col in df.select_dtypes(include=np.number).columns:
   df[col] = df[col].fillna(df[col].median())
for col in df.select_dtypes(include='object').columns:
 if not df[col].mode().empty:
   df[col] = df[col].fillna(df[col].mode()[0])
print("\nMissing Values After Cleaning:")
print(df.isnull().sum())
```

```
Missing Values After Cleaning:
YearsExperience    0
Salary             0
dtype: int64
```

In [13]:
```
columnname = 'Salary'
Q1 = df[columnname].quantile(0.25)
Q3 = df[columnname].quantile(0.75)
IQR = Q3 - Q1
lower = Q1 - 1.5 * IQR
upper = Q3 + 1.5 * IQR
print("\nQ1:", Q1)
print("Q3:", Q3)
print("IQR:", IQR)
print("Lower Limit:", lower)
print("Upper Limit:", upper)
# Find Outliers
outliers = df[
 (df[columnname] < lower) |
 (df[columnname] > upper)
]
print("\nOutliers:")
```

```
print(outliers)
print("Number of Outliers:", len(outliers))
```

```
Q1: 56720.75
Q3: 100544.75
IQR: 43824.0
Lower Limit: -9015.25
Upper Limit: 166280.75

Outliers:
Empty DataFrame
Columns: [YearsExperience, Salary]
Index: []
Number of Outliers: 0
```

In [17]:
```
df = df[
 (df[columnname] >= lower) &
 (df[columnname] <= upper)
].copy()
print("\nData After Removing Outliers:")
print(df.shape)
```

```
Data After Removing Outliers:
(30, 2)
```

In [18]:
```
X = df[['YearsExperience']]
y = df['Salary']
print("\nIndependent Variable (X):")
print(X)
print("\nDependent Variable (y):")
print(y)
```

```
Independent Variable (X):
    YearsExperience
0               1.1
1               1.3
2               1.5
3               2.0
4               2.2
5               2.9
6               3.0
7               3.2
8               3.2
9               3.7
10              3.9
11              4.0
12              4.0
13              4.1
14              4.5
15              4.9
16              5.1
17              5.3
18              5.9
19              6.0
20              6.8
21              7.1
22              7.9
23              8.2
24              8.7
25              9.0
26              9.5
27              9.6
28             10.3
29             10.5

Dependent Variable (y):
0      39343.0
1      46205.0
2      37731.0
3      43525.0
4      39891.0
5      56642.0
6      60150.0
7      54445.0
8      64445.0
9      57189.0
10     63218.0
11     55794.0
12     56957.0
13     57081.0
14     61111.0
15     67938.0
16     66029.0
17     83088.0
18     81363.0
19     93940.0
20     91738.0
21     98273.0
22    101302.0
23    113812.0
24    109431.0
25    105582.0
26    116969.0
27    112635.0
28    122391.0
29    121872.0
Name: Salary, dtype: float64
```

In [19]:

```python
X = df[['YearsExperience']]
y = df['Salary']
print("\nIndependent Variable (X):")
print(X)
print("\nDependent Variable (y):")
print(y)
```

```
Independent Variable (X):
    YearsExperience
0               1.1
1               1.3
2               1.5
3               2.0
4               2.2
5               2.9
6               3.0
7               3.2
8               3.2
9               3.7
10              3.9
11              4.0
12              4.0
13              4.1
14              4.5
15              4.9
16              5.1
17              5.3
18              5.9
19              6.0
20              6.8
21              7.1
22              7.9
23              8.2
24              8.7
25              9.0
26              9.5
27              9.6
28             10.3
29             10.5

Dependent Variable (y):
0      39343.0
1      46205.0
2      37731.0
3      43525.0
4      39891.0
5      56642.0
6      60150.0
7      54445.0
8      64445.0
9      57189.0
10     63218.0
11     55794.0
12     56957.0
13     57081.0
14     61111.0
15     67938.0
16     66029.0
17     83088.0
18     81363.0
19     93940.0
20     91738.0
21     98273.0
22    101302.0
23    113812.0
24    109431.0
25    105582.0
26    116969.0
27    112635.0
28    122391.0
29    121872.0
Name: Salary, dtype: float64
```

In [24]:
```
X_train, X_test, y_train, y_test = train_test_split(
 X,
 y,
 test_size=0.2,
 random_state=12
)
print("\nTraining Data:", X_train.shape)
print("Testing Data:", X_test.shape)
```

```
Training Data: (24, 1)
Testing Data: (6, 1)
```

In [29]:
```
import numpy as np
from sklearn.metrics import mean_absolute_error, mean_squared_error, r2_score
```

In [ ]:
```

```

In [ ]: